Scrap

Senior Design (423-IDE401 / 424-IDE402) Core Report

Version 1.5

Milestone 1 Draft

by

Team Scrap

Stevens Institute of Technology

September 10, 2026

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Template Attribution

This Senior Design Core Report template was adapted from the software architecture design template developed by **Darian Muresan, Ph.D.**, in conjunction with the AI agents **OpenAI Codex** and **Anthropic Claude Code**.

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Version 1.5
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Team Scrap
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This document specifies the stakeholder needs, requirements, architecture, implementation constraints, tests, verification evidence, validation evidence, and release criteria for the Scrap. It is both a printable engineering record and the authoritative input for AI-assisted implementation. A conforming implementation may be regenerated from the complete specification, but no single chapter should be treated as sufficient on its own.

Document History

Add the newest approved change at the top of the table. Retain every approved entry so that the report's evolution and review history remain auditable.

Table 2: Document Update History

Date / Version	Author	Change	Location
September 10, 2026 / 1.5	DMG, SAK, SAO	Made the Location column in this table clickable, linking each entry to its chapter or appendix. Added the GitHub repository setup to the Week 2 report.	Document History; Appendix A: Weekly Reports
September 10, 2026 / 1.4	DMG, SAK, SAO	Updated author initials throughout this table to reflect the full team. Clarified in the Week 2 report that the Key Drivers and Key Constraints chapters were started but not yet finished as of that week.	Document History; Appendix A: Weekly Reports
September 10, 2026 / 1.3	DMG, SAK, SAO	Added the Week 2 report to Appendix A, Weekly Reports. Removed the placeholder example rows from this table.	Appendix A: Weekly Reports; Document History
September 10, 2026 / 1.2	DMG, SAK, SAO	Added Appendix A, Weekly Reports (Week 1 report), and Appendix B, AI Prompt Log, to this Milestone 1 submission copy.	Appendix A: Weekly Reports; Appendix B: AI Prompt Log
September 10, 2026 / 1.1	DMG, SAK, SAO	Created this trimmed Milestone 1 submission copy from the full working template, containing only the front matter and the chapters completed to date.	Entire document

Continued on next page

Date / Version	Author	Change	Location
September 8, 2026 / 1.1	DMG, SAK, SAO	Customized the template for Scrap. Added the Team Declaration chapter (team name, members, project proposal) and the Milestone 1 content: Mission Statement, Key Drivers, and Key Constraints. Updated project metadata.	Team Declaration; Mission Statement; Key Drivers; Key Constraints
September 1, 2026 / 1.0	DDM, CDX	Adapted the system and software architecture design template as the Senior Design Core Report template. Added the complete 423-IDE401 / 424-IDE402 report structure, milestones, and appendices.	Entire report

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Part I

Project Definition and Plan

Chapter 1

Team Declaration

1.1 Team Name

Scrap

1.2 Team Members

- **Salome Orozco** is a fourth-year Industrial and Systems Engineering student at Stevens Institute of Technology, pursuing a minor in Finance. Originally from Texas, she came to Stevens to challenge herself and grow independently, choosing a course of study that blends the technical side of engineering with the strategic side of business. She currently serves as President of the First-Generation Student Union and thrives on hands-on projects that solve complex, structural problems.
- **Samah Keshiro** is a fourth-year Software Engineering major at Stevens Institute of Technology with interests in Product Management and AI. Serving on a club executive board revealed her strength for identifying requirements and managing timely delivery, and she is now working toward a career as a Technical Product Manager for software and digital products, building on AI engineering experience gained at Quest Diagnostics.
- **Deanna Gaber** is a fourth-year Software Engineering major at Stevens Institute of Technology focused on software development, systems design, and improving efficiency through technology-driven solutions. She values collaborative environments where ideas and responsibilities are shared and is using her coursework to strengthen her programming skills and prepare for a career in the technology industry.

1.3 Project Proposal

Our team, Scrap, proposes to develop a recipe and meal-planning application designed to help users make better use of the food they already have while reducing unnecessary food waste. The application will allow users to enter ingredients they currently have available and

receive recipe recommendations based on those ingredients. Rather than requiring users to shop for an entirely new set of ingredients for every recipe, the system will prioritize meals that make the greatest use of what is already in the user's kitchen.

The application will be centered around a simple search experience where users can search for ingredients and apply additional filters to personalize their results. Recipes will be ranked according to the percentage of ingredients the user already has, allowing users to quickly identify meals that require the fewest additional purchases. When additional ingredients are needed, users will be able to compile those items into a grocery list. The system will also allow users to contribute new recipes, creating the potential for the application to become a more social and collaborative platform over time.

Chapter 2

Mission Statement

The mission of Scrap is to help people decide what to cook by maximizing the use of ingredients they already have while reducing unnecessary food waste. The application aims to make meal planning more affordable, convenient, and enjoyable by connecting users with recipes that fit their available ingredients, budget, and preferences.

Chapter 3

Key Drivers

The primary motivation for Scrap comes from our experiences as college students transitioning into apartments and becoming responsible for cooking and grocery shopping on our own. Cooking regularly can be challenging when balancing classes, work, social responsibilities, and other commitments. For students who are still developing their cooking skills, deciding what to make can become an additional source of stress, particularly when there are only a limited number of ingredients available.

Budget is another major driver behind the proposed solution. College students often have limited disposable income and need to be thoughtful about how they spend money on groceries. Many existing recipes assume that users have access to a large variety of ingredients or require purchasing several items that may only be used once. This can make following a recipe more expensive than expected and can leave users with partially used ingredients that eventually expire. For students living alone or with a small number of roommates, this problem is especially noticeable because purchasing larger quantities of food does not always make sense for smaller portions.

Another challenge we identified is the difficulty of finding recipes that accommodate what a person already has in their kitchen. Traditional recipe searches typically begin with a desired meal, causing users to search for a recipe first and then purchase the ingredients required to make it. Scrap approaches this problem from the opposite direction: users begin with what they already have. By entering their available ingredients, users can discover recipes that make use of those items and determine which additional ingredients, if any, they need to purchase.

Food waste is also an important motivation for the project. Ingredients are frequently purchased with good intentions but are forgotten about or left unused until they expire. Scrap is intended to encourage users to think about their existing food inventory before purchasing additional groceries. Ranking recipes based on the percentage of ingredients a user already has can help users prioritize meals that use ingredients that might otherwise go to waste. In this way, the application connects personal benefits, such as saving money and reducing grocery trips, with a broader goal of reducing unnecessary food waste.

Finally, we want Scrap to be more than a basic recipe search engine. We envision the application as a social and engaging platform that makes deciding what to cook more enjoyable. Users will be able to search for recipes, explore recommendations, create grocery lists,

and contribute their own recipes to the system. This user-driven approach could allow the recipe collection to grow while giving users a sense of participation in the platform.

Chapter 4

Key Constraints

One of the major constraints for Scrap is the existing competition in the food and recipe application market. There are already many established applications and websites that provide recipe searches, meal planning, grocery lists, ingredient-based searches, and other related features. Because of this, simply creating another recipe application may not provide enough differentiation. Our project will need to clearly define what makes Scrap useful and distinct, particularly its focus on maximizing ingredients that users already have and reducing food waste. The interface and functionality will need to support this central idea rather than simply reproducing features that are already common in existing applications.

Another significant constraint is our team's development experience. While all three team members have experience developing websites and working with software development tools, we have not previously built a complete application of this type. Developing Scrap will therefore require us to learn and apply new technologies, development practices, and potentially unfamiliar frameworks. We will need to account for this learning curve when determining the scope of the project and deciding which features are realistic to implement within the available time.

The project is also constrained by the need to balance functionality with development complexity. Our initial vision includes ingredient-based recipe searches, filters, recipe rankings, grocery lists, and user-submitted recipes. Although these features would make the application more useful, implementing all of them successfully may require significant time and technical effort. We will need to prioritize the core functionality of the application and distinguish between essential features for the minimum viable product and features that could be added if time permits.

Data management presents another potential constraint. The application will depend on a collection of recipes and their associated ingredients in order to provide meaningful recommendations. We will need to determine how recipes are stored, how ingredients are standardized, and how the system handles differences in ingredient names, quantities, and measurements. For example, a user's available ingredient may be entered differently from how that ingredient appears in a recipe. The system will need to account for these differences to ensure that recipe recommendations are accurate and useful.

Finally, usability is an important constraint because the application is intended to be simple and convenient for users who may already be short on time. The core experience

should allow users to quickly enter what they have and receive useful recommendations without requiring extensive setup. As we develop the application, we will need to balance the number of available filters and features with the simplicity of the interface. The final product should demonstrate the core concept of Scrap clearly while remaining achievable within the technical and time limitations of the project.

Part II

Appendices

Appendix A

Weekly Reports

Add the newest week's report as a new section at the end of this chapter. Retain every prior week's report so that the team's weekly progress remains a complete, auditable record.

A.1 Week Report 1 (September 1, 2026)

1. What We Did This Past Week

This week, our group was formed and we selected an initial project idea. We discussed our interests and goals as a team and worked together to determine a direction for our project. We also began organizing our team and establishing a foundation for the work we will complete throughout the semester.

2. What We Will Do Next Week

Next week, we will conduct market research to gain a better understanding of our target market and the products or services related to our project idea. We will use this research to better define our product's goals, identify potential users, and understand the needs our product should address.

3. Action Items

- Conduct initial market research.
- Research similar products and competitors.
- Identify the target users for our product.
- Further define the goals and purpose of our product.
- Continue setting up our project documentation in Overleaf.

4. Issues and Risks

One issue our team is currently facing is that using LaTeX and Overleaf is new to one of our team members. To address this, the team will work together to help everyone become comfortable with the tools and ensure that all members can contribute effectively to the project documentation. We will also communicate regularly and provide assistance when needed to prevent this from becoming a larger issue as the project progresses.

A.2 Week Report 2 (September 8, 2026)

1. What We Did This Past Week

This week, we used AI to help compile and format our project report. We finished our Team Declaration and Mission Statement chapters and started drafting our Key Drivers and Key Constraints chapters, continued setting up our project documentation, and set up our GitHub repository. This work is not yet finished, and we are still working toward completing our Milestone 0 report.

2. What We Will Do Next Week

Next week, we plan to conduct more specific market research, including writing out the features we like and the features we find missing from competing products. We were not able to complete this market research last week because we spent more of our time on setting up our project and documentation. Next week, we also each plan to conduct an interview with someone outside our team to gauge whether there is demand for our product.

3. Action Items

- Conduct specific market research on competing recipe and meal-planning applications.
- Document the features we like and the features that are missing from competitor products.
- Each team member conducts one interview to assess demand for our product.

4. Issues and Risks

Because we spent more time on project setup this past week, our planned market research was delayed by one week. To stay on schedule, we will need to complete both the delayed market research and the new interviews next week, which increases our workload for that week.

Appendix B

AI Prompt Log

B.1 How to Use This Appendix

Use this log when AI assistance materially affects requirements, architecture, code, tests, or release evidence. Record enough context for a reviewer to understand the input, output, assumptions, and human corrections.

B.2 Log Entries

- **Date:** September 8, 2026. **Tool and model:** Claude (Cowork), model claude-sonnet-5. **Input:** the team's project proposal for Scrap (an ingredient-first recipe and meal-planning app), the team members' self-written biographies, and the team's own drafted Mission Statement, Key Drivers, and Key Constraints text. **Task:** add a new Team Declaration chapter to the report template (team name, shortened member bios, project proposal, within a two-page limit) and populate the existing Mission Statement chapter plus two new chapters, Key Drivers and Key Constraints, with the team-supplied content, following the template's authoring conventions. **Human review and corrections:** the team supplied all substantive content (proposal text, mission statement, drivers, constraints, and bios); the team requested the bios be shortened, which the AI did; the compiled PDF was reviewed by the team for page limits and formatting before acceptance. **Unresolved assumptions:** none.
- **Date:** September 10, 2026. **Tool and model:** Claude (Cowork), model claude-sonnet-5. **Input:** the full working report template as customized on September 8, 2026. **Task:** produce a separate, trimmed LaTeX project (`Scrap-Milestone1`) containing only the front matter and the chapters completed so far (Team Declaration, Mission Statement, Key Drivers, Key Constraints), suitable for a Milestone 1 submission, delivered as source files rather than a compiled PDF. **Human review and corrections:** none required; the AI compiled the trimmed project and reported the result (page count, table of contents) for the team to confirm before delivery. **Unresolved assumptions:** whether this trimmed submission copy should be refreshed automatically each week as new chapters are added, or rebuilt only when a submission is due, is still open.

- **Date:** September 10, 2026. **Tool and model:** Claude (Cowork), model claude-sonnet-5. **Input:** the team's Week 1 report text (accomplishments, next week's plan, action items, issues and risks), supplied verbatim by the team. **Task:** add a new Appendix A, Weekly Reports, to the report (both the full template and this Milestone 1 submission copy), formatted as one section per week with the team's four-part structure, and record this and prior AI-assisted changes in this AI Prompt Log. **Human review and corrections:** the team supplied the report text verbatim; the AI only handled formatting, chapter placement (ensuring it renders as Appendix A), and the corresponding Document History entries. **Unresolved assumptions:** none.

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[3] *ISO/IEC/IEEE 29148:2018, Systems and Software Engineering—Life Cycle Processes—Requirements Engineering*, International Organization for Standardization Std. ISO/IEC/IEEE 29 148:2018, 2018. [Online]. Available: https://www.iso.org/standard/72089.html

Glossary

acceptance criterion An observable condition used to decide whether a need, story, or requirement has been satisfied. 14

Could A desirable capability included only when time and resources permit. 14

Must A capability required in the current release; the release is unacceptable without it. 14

requirement A necessary, singular, unambiguous, feasible, and verifiable statement of capability or constraint. 14

Should An important capability expected in the current release unless a documented constraint prevents it. 14

stakeholder A person, group, or organization that affects the system or is affected by it. 14

traceability The ability to follow an artifact backward to its source and forward to its implementation and evidence. 14

use case A goal-oriented interaction between an actor and the system, including normal and alternate flows. 14

V-model A lifecycle model that pairs each decomposition and specification activity with a corresponding integration and evidence activity. 14

validation Confirmation that the completed system satisfies stakeholder needs in its intended context; building the right system. 14

verification Confirmation that specified requirements have been fulfilled; building the system right. 14

Would A capability assigned to a later implementation version after higher-priority work. 14

Acronyms

AI artificial intelligence. 14

UI user interface. 14

UML Unified Modeling Language. 14

V&V verification and validation. 14

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